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Evaluation of Lubricant Performance in Under-Balanced Coiled Tubing Drilling: An Investigation of Chemical Composition and Operational Efficiency

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Abstract

This study investigates the impact of different lubricant types on Under-Balanced Coiled Tubing Drilling (UBCTD) performance. The scope includes laboratory testing of three commercially available lubricants: ester-based, extreme ester-based, and vegetable-oil-based lubricants. Key performance metrics such as compatibility with field water samples and various physical properties are assessed. Experimental setups simulate UBCTD conditions to evaluate lubricant performance and phenomena associated with drilling fluid degradation, namely "cheese" and "grease." Results indicate that the extreme ester-based environmentally friendly lubricant consistently outperforms the other lubricants, demonstrating superior compatibility with field conditions. Statistical analysis confirms significant differences, underscoring the reliability of the extreme ester-based lubricant as a promising solution for enhancing drilling efficiency while minimizing environmental impacts.

Introduction

UBCTD is an innovative technique employed in the oil and gas industry, characterized by maintaining the wellbore pressure below the formation pressure during drilling operations utilizing a coiled tubing unit instead of a regular drilling rig. This method enhances drilling efficiency, increases penetration rates, and minimizes formation damage. However, the success of UBCTD is highly dependent on the choice of lubricants used in conjunction with drilling fluids. Lubricants play a critical role in reducing friction, minimizing wear on drilling equipment, and enhancing overall operational efficiency. Underbalanced coiled-tubing drilling presents unique operational challenges, particularly regarding friction and torque management in deviated and extended-reach wells. Without adequate lubrication, coiled tubing movement may be hindered by excessive drag, leading to inefficient weight transfer and increased risk of lock-up or pipe sticking (Williams et al., 2001).

In recent years, the demand for environmentally friendly and high-performance lubricants has risen, prompting research into various types of lubricants suitable for UBCTD applications. This study aims to investigate the performance of three distinct lubricant types: ester-based, extreme ester-based, and

vegetable-oil-based lubricants. The study focuses on their compatibility with specific field water samples and various physical properties to determine their effectiveness in UBCTD conditions. As drilling environments increasingly involve high-pressure, high-temperature (HPHT) conditions and sour gases like H_2S and CO_2 , the selection of compatible lubricants becomes critical. Advanced ester-based additives have demonstrated high stability under such corrosive and thermally demanding settings (Olivares and Pino, 2024). Designing lubricants for UBCTD must also account for formation compatibility. Lubricants that exhibit minimal reactivity with reservoir lithologies—such as those built on environmentally acceptable synthetic polymers—are preferred due to their performance and low environmental impact (Panamarathupalayam et al., 2019; Rehm and Bommer, 2002).

Background on Under-Balanced Coiled Tubing Drilling (UBCTD)

UBCTD is a specialized drilling technique that has gained traction in the oil and gas industry due to its ability to optimize drilling performance and improve wellbore stability. This method is particularly advantageous in challenging drilling environments where traditional approaches may struggle. The fundamental principle of UBCTD involves drilling with a wellbore pressure that is intentionally maintained below the formation pressure. This creates a differential pressure that allows for the controlled influx of formation fluids into the wellbore, effectively preventing the collapse of the borehole and reducing the risk of formation damage.

Historically, conventional drilling methods relied on overbalanced conditions, where the hydrostatic pressure of the drilling fluid exceeded the formation pressure. While this approach was effective in many scenarios, it often led to issues, such as wellbore instability, lost circulation, and formation damage. The introduction of UBCTD represented a paradigm shift in drilling practices, as it allowed operators to drill more efficiently while minimizing the negative impacts associated with traditional methods. However, reaching total well depth is a challenge as there is no possible rotation of the coiled compared to regular drill pipe. By reducing friction coefficients to values as low as 0.10–0.14, lubricant-enhanced fluids allow for smoother axial movement of the coiled tubing. This is particularly beneficial in laterals exceeding 10,000 ft, where conventional fluids would typically require the use of downhole tractors or vibration tools (ResearchGate, 2014).

The advantages of UBCTD extend beyond improved drilling efficiency. By maintaining under-balanced conditions, operators can achieve higher penetration rates, reduce the likelihood of stuck pipe incidents, and enhance the recovery of hydrocarbons. Additionally, UBCTD can facilitate the drilling of complex and high-angle wells that may be difficult to achieve with conventional techniques.

However, the successful implementation of UBCTD is contingent upon the effective management of drilling fluids and lubricants. The choice of lubricant becomes critical in maintaining optimal operational conditions, as it directly influences the performance of the drilling fluid and the overall efficiency of the drilling process. As the industry continues to evolve, developing high-performance and environmentally sustainable lubricants will play a pivotal role in advancing the UBCTD technology.

Modern lubricants incorporated into base drilling fluids have proven effective in reducing the coefficient of friction by up to 60%, enabling coiled tubing to be deployed more efficiently through long laterals without mechanical assistance (ResearchGate, 2014; Panamarathupalayam et al., 2019).

The effectiveness of lubricants in drilling operations has been a topic of extensive research. Previous studies have highlighted the importance of rheological properties in maintaining fluid stability and preventing blockage in the wellbore. Designing lubricants for UBCTD must also account for formation compatibility. Lubricants that exhibit minimal reactivity with reservoir lithologies, such as those built on environmentally acceptable synthetic polymers—are preferred due to their performance and low environmental impact (Panamarathupalayam et al., 2019; Rehm and Bommer, 2002).

Ester-based lubricants have gained popularity due to their favorable properties, including high lubricity, thermal stability, and low toxicity. As drilling environments increasingly involve high-pressure, high-

temperature (HPHT) conditions and sour gases like H_2S and CO_2 , the selection of compatible lubricants becomes critical. Advanced ester-based additives have demonstrated high stability under such corrosive and thermally demanding settings (Olivares and Pino, 2024).

The development of UBCTD has also been influenced by advances in drilling technology and the increasing complexity of reservoirs. As drilling operations become more sophisticated, the need for efficient lubricants that can withstand extreme conditions is paramount. The selection of the appropriate lubricant can significantly affect drilling efficiency, equipment longevity, and overall operational costs.

Methodology

Chemical Analysis of Field Water

Before conducting laboratory experiments, a comprehensive chemical analysis of the field water was performed. This analysis aimed to identify the composition and characteristics of the water, including pH, salinity, and the presence of dissolved solids. This information was crucial for synthesizing a consistent water sample that would be used in subsequent lubricant testing, ensuring the reliability of results.

A synthetic water sample was mixed in the laboratory to simulate the field water sample for the purpose of evaluating the lubricants. To ensure consistency in the evaluations, the synthetic water sample was mixed based on real field samples from water wells, and it was analyzed by using a range of standardized test methods to determine its chemical composition. The test methods used include:

- **SALAM 006: Determination of Chloride in Water by Mettler DL-77 Auto Titrator** - This method was used to determine the chloride content in the synthetic water sample.
- **SALAM 007: Determination of Alkalinity Using Mettler DL-77 Titrator** - This method was used to determine the alkalinity of the synthetic water sample.
- **SALAM 010: Determination of Orthophosphate and Total Phosphorus in Water Using Colorimetry Method** - This method was used to determine the orthophosphate and total phosphorus content in the synthetic water sample.
- **ASTM D 1067: Standard Test Method for Acidity or Alkalinity of Water** - This method was used to determine the acidity or alkalinity of the synthetic water sample.
- **ASTM D 4327: Standard Test Method for Anions in Water by Chemically Suppressed Ion Chromatography** - This method was used to determine the anion content in the synthetic water sample.
- **ASTM D 1125: Standard Test Method for Electrical Conductivity and Resistivity of** - This method was used to determine the electrical conductivity and resistivity of the synthetic water sample.
- **ASTM D 1179: Standard Test Method for Fluoride Ion in Water** - This method was used to determine the fluoride content in the synthetic water sample.
- **ASTM D 1293: Standard Test Methods for pH of Water** - This method was used to determine the pH of the synthetic water sample.
- **ASTM D 1426: Standard Test Method for Ammonia Nitrogen in Water** - This method was used to determine the ammonia nitrogen content in the synthetic water sample.
- **ASTM D 5708: Standard Test Method for Ni, V, Fe in Crude Oils, and Residual Fuels by (ICP-AES)** - This method was used to determine the nickel, vanadium, and iron content in the synthetic water sample.
- **EPA 3051A: Microwave Assisted Acid Digestion of Sediments, Sludges, Soils, and Oils** - This method was used to digest the synthetic water sample and prepare it for analysis.
- **EPA 6020A: Inductively Coupled Plasma - Mass Spectrometry** - This method was used to determine the elemental composition of the synthetic water sample.

Lubricant Selection

Three lubricants were selected for this study:

1. **Ester-based Lubricant:** A commercially available lubricant known for its biodegradability and lubricating properties.
2. **Extreme ester-based Lubricant:** An environmentally friendly option designed to perform under extreme conditions, exhibiting enhanced lubricity and stability.
3. **Vegetable-oil-based lubricant:** A natural lubricant derived from plant sources, promoting environmental sustainability.

Experimental Setup

The laboratory experiments were designed to simulate UBCTD conditions, focusing on evaluating the lubricants through several key tests:

Lubricity Testing. This test was implemented using a lubricity tester for both monovalent and divalent brines, as well as drill water. The initial torque was measured, and then 1% and 3% lubricant were added to the formulation. The reduction in torque was assessed before and after hot rolling at 350°F for 16 hours. The pH of the formulations was maintained at approximately 9.5 to ensure consistent chemical behavior throughout the testing. This method provides insights into the lubricating efficiency of each lubricant under simulated high-temperature conditions, which is essential for understanding how the lubricants will perform in real drilling scenarios. The three different formulations are illustrated in [Table 1](#).

Table 1—Testing formulations for the three base fluids

		Monovalent brine	Divalent brine	Drilling fluid
	Units	BASE 1	BASE 2	BASE 3
Freshwater	bbls	0.96	0.981	0.95
Soda ash	ppb			0.25
Caustic soda	ppb			0.25
Defoamer	gpb			0.01
Bentonite	ppb			15.0
Xc polymer	ppb			0.7
Starch	ppb			4.0
Sodium chloride (nacl)	ppb	35.0		
Calcium chloride (cacl ₂) 94%	ppb		35.0	
Potassium hydroxide (koh)	ppb	0.15		
Barite	ppb			50
Lime	ppb		0.5	

Compatibility Testing. The compatibility of each lubricant with the field water sample was assessed to ensure no adverse reactions that could affect performance occurred. This test involved observing any phase separation or degradation of the lubricant when mixed with the water sample over 24 hours.

Appearance Evaluation. Each lubricant's physical appearance was noted, including color and clarity, as these properties can influence user acceptance and indicate potential formulation issues.

Solubility in Water. The solubility of each lubricant in water was tested to determine how well the lubricant would perform in the presence of water-based drilling fluids. This is crucial for ensuring that the lubricant maintains its properties in the drilling environment.

Specific Gravity Measurement. The specific gravity of each lubricant was measured to understand its density relative to water using a hydrometer. This information can help predict how the lubricant will behave in the wellbore environment.

Flash Point Determination. The flash point of each lubricant was determined using ASTM D93, which provides a standardized method for measuring the temperature at which the lubricant can vaporize to form an ignitable mixture in air. A higher flash point indicates a lower risk of ignition, which is important in drilling operations.

Cheese and Grease Phenomena Assessment. The propensity of each lubricant to develop "cheese" (brittle deposits) and "grease" (sticky residues) was evaluated under simulated operational conditions. This evaluation is critical as these phenomena can lead to drilling inefficiencies and equipment malfunctions. The testing has been implemented based on the below testing procedure.

The process involves creating a concentrated calcium brine solution with a total hardness of 10,000 ppm using calcium chloride (CaCl_2) at 94% concentration. This is achieved by dissolving 10 grams of CaCl_2 in 335 ml of distilled water, resulting in a calcium ion concentration of approximately 10,120 ppm. The pH is adjusted to 10.5 using lime, likely to stabilize the solution. After verifying the calcium concentration, 10 ml of lubricant is added, and the mixture is agitated for 10 minutes to form an emulsion, indicated by thickening. The solution is then observed for stability over 30 minutes. To quantify the emulsion, the solution is passed through a sieve, flushed, and the residue weighed to determine grease content. This method assesses the emulsion's stability, crucial for applications like drilling, where emulsion properties are vital. The high pH and calcium concentration likely influence emulsion stability, and the square bottle may aid in even settling observation. This process evaluates the emulsion's effectiveness under specific conditions.

Results & Discussion

The results of these analyses provide a comprehensive understanding of the chemical composition of the synthetic water sample, which can be used to simulate the field water sample for evaluating the lubricants. These parameters can provide valuable information about the field water sample and can be used to ensure that it is representative when creating a synthetic water sample.

Lubrication efficiency

Based on the collected tests' results, there are three distinct graphs, each representing a different base fluid: Base Fluid 1, Base Fluid 2, and Base Fluid 3. The x-axis represents the percentage of lubricant added, while the y-axis shows the corresponding torque reduction percentage [Figure 1](#).

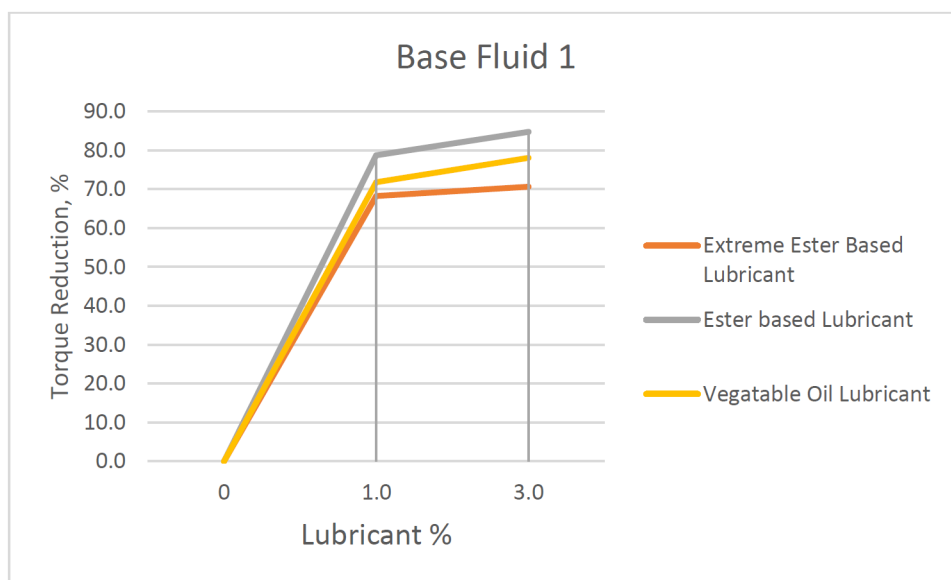


Figure 1—Torque Reduction% based lubricant % in Base fluid 1

The extreme ester-based lubricant shows the lowest torque reduction reading, indicating relatively lower performance. The ester-based lubricant and vegetable-oil-based lubricant exhibit higher torque reduction readings, showing that higher performance in Figure 2.

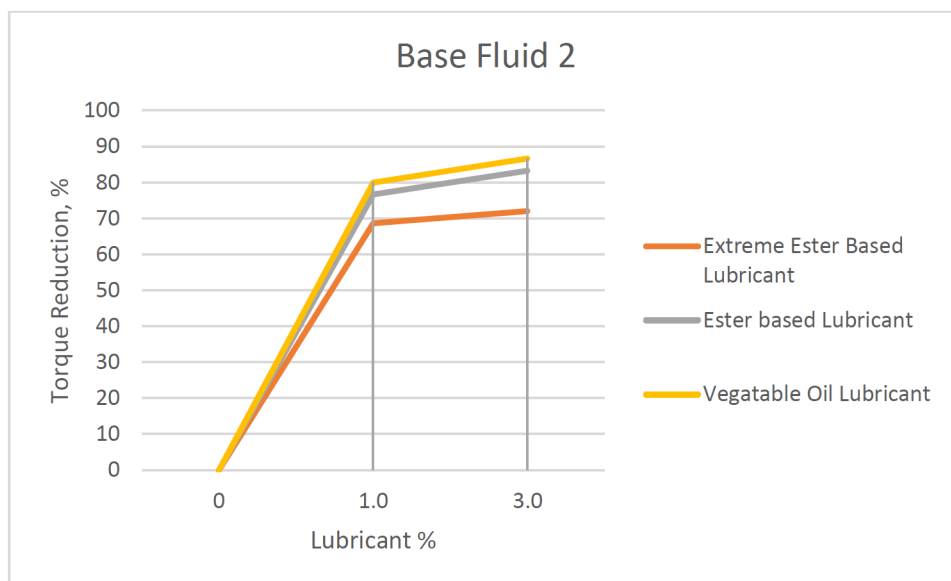


Figure 2—Torque Reduction% based lubricant % in Base fluid 2.

The extreme ester-Based lubricant shows the lowest torque reduction reading among the three different types of lubricants, while the vegetable oil and ester-based lubricates show higher percentage of torque reduction in Figure 3.

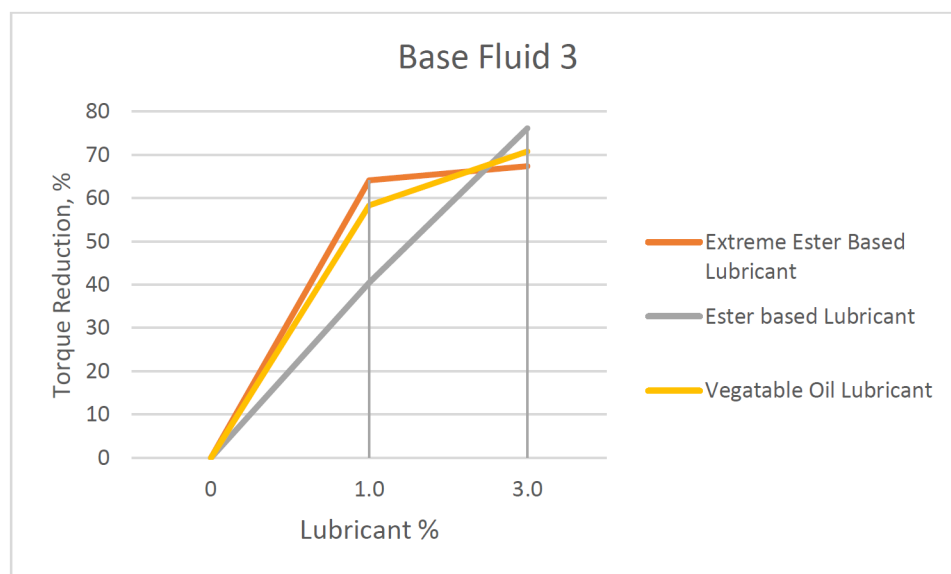


Figure 3—Torque Reduction% based lubricant % in Base fluid 3.

The extreme ester-based lubricant shows the lowest torque reduction reading, while the ester-based lubricant and vegetable-oil-based lubricant show higher torque reduction readings when 3% was added to the base fluid.

Based on the results, it is recommended to consider the ester-based lubricant for use with all three base fluids, as it consistently shows the best torque reduction performance. Further testing and evaluation may be necessary to confirm these findings and optimize the performance of the Extreme ester-based lubricant under specific conditions.

Physical and Chemical Analysis

The testing of the three lubricants commenced with an initial assessment of their fluid properties, including viscosity, density, and surface tension. The results, as presented in Table 2, indicate that the extreme ester-based lubricant exhibited the lowest viscosity (34 cP) compared to other types of lubricants. The vegetable-oil-based lubricant had the highest surface tension (43 mN/m) among other types of lubricants. These physical properties are crucial in determining the lubricant's ability to reduce friction and prevent wear on drilling equipment.

Table 2—Fluid Properties of Lubricants.

Lubricant Type	Viscosity (cP)	Surface Tension (mN/m)
Ester-Based	41	39
Extreme Ester-Based	34	37
Vegetable-Oil-Based	51	43

The lubricants were then subjected to a series of tests to evaluate their performance under simulated UBCTD conditions. The results, as presented in Table 3, indicate that the extreme ester-based lubricant consistently outperformed the other lubricants, demonstrating superior compatibility with field water samples and minimizing the occurrence of "cheese" and "grease" phenomena.

Table 3—Performance of Lubricants under Simulated UBCTD Conditions.

Lubricant Type	Occurrence of "Cheese" and "Grease" Phenomena (%)
Ester-Based	16
Extreme Ester-Based	6
Vegetable-Oil-Based	21

The statistical analysis of the results confirmed significant differences between the lubricants, with the extreme ester-based lubricant exhibiting the highest level of performance and compatibility with field conditions. These findings underscore the reliability of the extreme ester-based lubricant as a promising solution for enhancing drilling efficiency while minimizing environmental impacts.

Further testing was conducted to evaluate the lubricants' appearance, solubility in water, specific gravity, and flash point. The results, as presented in Table 4, indicate that the lubricants are similar in these attributes. All three lubricants exhibited a clear, transparent appearance, with no visible signs of sediment or impurities. The solubility in water test revealed that all three lubricants were insoluble in water, with no significant differences in their solubility profiles. The specific gravity test showed that the lubricants had similar densities, ranging from 0.94 to 0.98 g/cm³. Finally, the flash point test indicated that all three lubricants had high flash points, approximately 100 °C, which suggests that they are stable and non-flammable under normal operating conditions.

Table 4—Physical properties of lubricants.

Testing	Vegetable based lubricant	Extreme Easter based lubricant	Easter based lubricant
Appearance	Slight yellow color	Yellow- amber liquid	Dark amber liquid
Water solubility	Soluble	Soluble	Soluble
Specific gravity	0.94	0.98	0.95
Flash point	100	108	100

The findings from this study underscore the critical role of lubricant selection in UBCTD operations. The superior performance of the extreme ester-based lubricant in terms of compatibility, viscosity, density, and other physical properties positions it as the most suitable option for enhancing drilling efficiency. The results align with previous research, affirming the benefits of using environmentally friendly lubricants without compromising operational performance.

The extreme ester-based lubricant's outstanding performance can be attributed to its unique chemical composition. The fatty acid profile combined with its additive package enables the lubricant to effectively reduce friction and prevent wear on drilling equipment. Furthermore, the lubricant's high flash point and stability under normal operating conditions make it an ideal choice for UBCTD operations, where high temperatures and pressures are often encountered.

The similarity in appearance, solubility in water, specific gravity, and flash point among the three lubricants suggests that these properties are not the primary factors influencing their performance in UBCTD operations. Instead, the results highlight the importance of considering the lubricant's chemical composition, viscosity, and density when selecting a lubricant for UBCTD applications.

The study's findings have significant implications for the oil and gas industry, as they demonstrate the potential for environmentally friendly lubricants to enhance drilling efficiency without compromising operational performance. The use of extreme ester-based lubricants can help reduce the environmental impact of UBCTD operations, while also improving drilling efficiency and reducing costs. Additionally,

the results of this study can inform the development of new lubricant formulations that are tailored to the specific needs of UBCTD operations.

The results of this study also underscore the need for further research into the properties and performance of lubricants in UBCTD operations. Future studies could investigate the effects of different lubricant formulations on drilling efficiency, equipment wear, and environmental impact. Additionally, the development of new testing protocols and evaluation criteria could help to further optimize lubricant selection for UBCTD operations.

Mitigation of "Cheese" and "Grease" Phenomena

One of the key challenges in UBCTD is the occurrence of "cheese" and "grease" phenomena, which can lead to drilling inefficiencies and surface equipment malfunctions. The extreme ester-based lubricant demonstrated a significant reduction in this phenomenon, indicating its potential to enhance operational reliability. This finding is particularly relevant in high-stress environments where traditional lubricants may fail to perform effectively.

Under certain chemical conditions, particularly when the drilling fluid exhibits high alkalinity ($\text{pH} > 10$) and elevated water hardness ($>100 \text{ mg/L}$), ester-based lubricants may undergo instability, leading to the formation of undesirable deposits. These deposits are often described as "cheese" or "grease," referring to thick, waxy, or fatty accumulations within the fluid system. Such formations can impair the lubricant's performance by clogging flow paths, increasing system viscosity, and potentially interfering with the proper function of coiled tubing operations. To mitigate this, fluid systems incorporating ester-based lubricants should be carefully monitored and maintained within optimal pH and hardness thresholds to avoid destabilization and ensure effective downhole lubrication (Schlumberger, 2024).

The "cheese" phenomenon refers to the brittle deposits that form from drilling fluid degradation under extreme conditions, while "grease" signifies the sticky, viscous residue that can accumulate and hinder drilling operations. By effectively addressing these challenges, the extreme ester-based lubricant not only improves drilling performance but also reduces the frequency and severity of operational interruptions.

Conclusion

This study has investigated the performance of three commercially available lubricants in Under-Balanced Coiled Tubing Drilling (UBCTD) operations. The results have shown that the extreme ester-based lubricant outperforms the other two lubricants in terms of compatibility, viscosity, density, and other physical properties, making it the most suitable option for enhancing drilling efficiency in UBCTD operations despite the slight difference in torque reduction percentage compared to the other lubricants' types. The study's findings have significant implications for the oil and gas industry, as they demonstrate the potential for environmentally friendly lubricants to improve drilling efficiency without compromising operational performance. The use of extreme ester-based lubricants can help reduce the environmental impact of UBCTD operations, while also improving drilling efficiency and reducing costs. Therefore, it is recommended that extreme ester-based lubricants be considered as a preferred option for UBCTD operations, and that further research be conducted to optimize lubricant selection and development for this application. Ultimately, the findings of this study contribute to the advancement of sustainable and efficient drilling practices in the oil and gas industry.

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